VIPPS · CASE INTERVIEW

Does VIP create customer loyalty?

Assessing VIP's impact on spending, retention, and engagement.

419	350,567	30	269
Users in the dataset	Purchases	Merchants	Experiment users

01 · DEFINE

How do we measure loyalty?

Define "Keep coming back" in a way that reflects how customers use the service.

02 · EVALUATE

Does VIP change behaviour?

Compare spending, retention, and engagement across VIP and non VIP users, experiment & control groups.

03 · RECOMMEND

What should happen next?

Turn the findings into a decision, follow-up analysis, and success measures.

- From defining loyalty to evaluating impact and recommending action

Repository ↗

EXECUTIVE SUMMARY

Invites drive signups, not retention.

Customers don't gradually slow down: They either use us multiple times a day or stop completely (~29% churn before year-end).

01 · DEFINE SUCCESS

Keep coming back

We define repeat engagement through three metrics:

- **Survival:** The user made a purchase in the last **28** days.
- **Cadence:** The user maintains the expected purchase frequency.
- **Spend:** The user continues to make purchases and generate GMV.

Together, these metrics capture whether users remain active, return regularly, and continue spending on the platform.

02 · EARLY VIP ADOPTERS

Customers who adopted VIP were already more active

Future VIP customers spent **1.09×** more per week before launch. After enrollment, purchases rose **8.4%** and GMV **5.5%** within user, driven by in-person activity.

03 · INVITATION EXPERIMENT

Experiment boosts signup, not retention

Signup increased from **8.0%** to **68.1%** **(+60.2 pp)** (**p < 0.001**). However, it produced no statistically significant lift in retention or spend (**4.7%** purchases , **-0.5%** GMV).

04 · WHAT TO DO NEXT

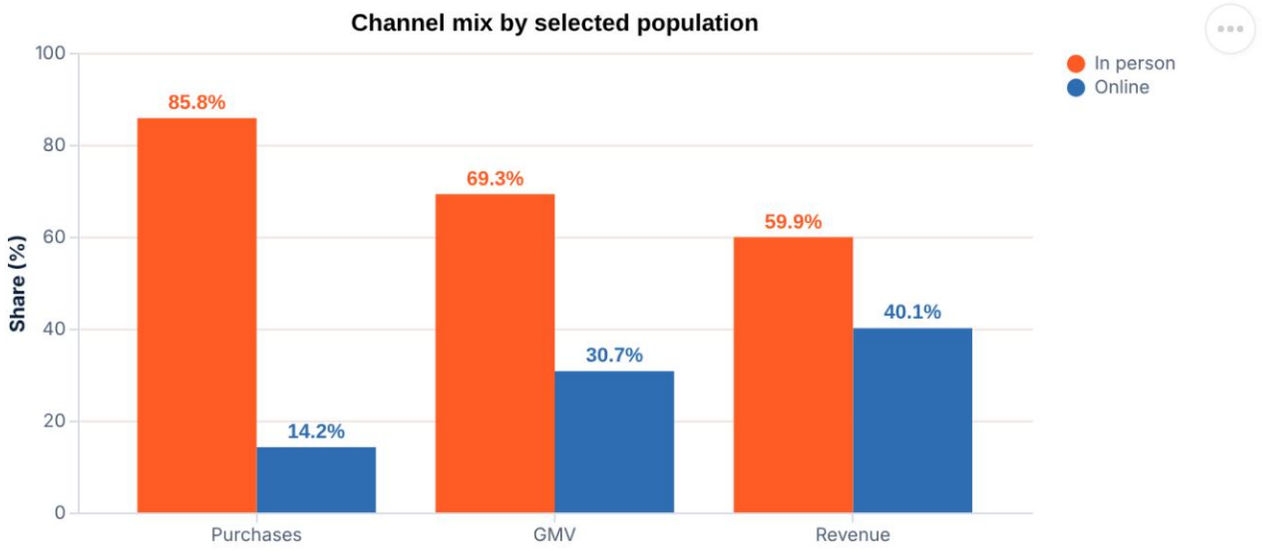
Passive signup nudges aren't enough

Test 14-day inactivity reminders. Online checkout point multipliers (Online take rate is **1.95%** versus **1.29%** for in-person).

BOTTOM LINE Invitations are effective at converting users into VIPs; retention impact remains unproven.

CHANNEL ECONOMICS

Two distinct products in one: daily habit and high-value retail



CHANNEL COMPARISON

Two distinct spending patterns

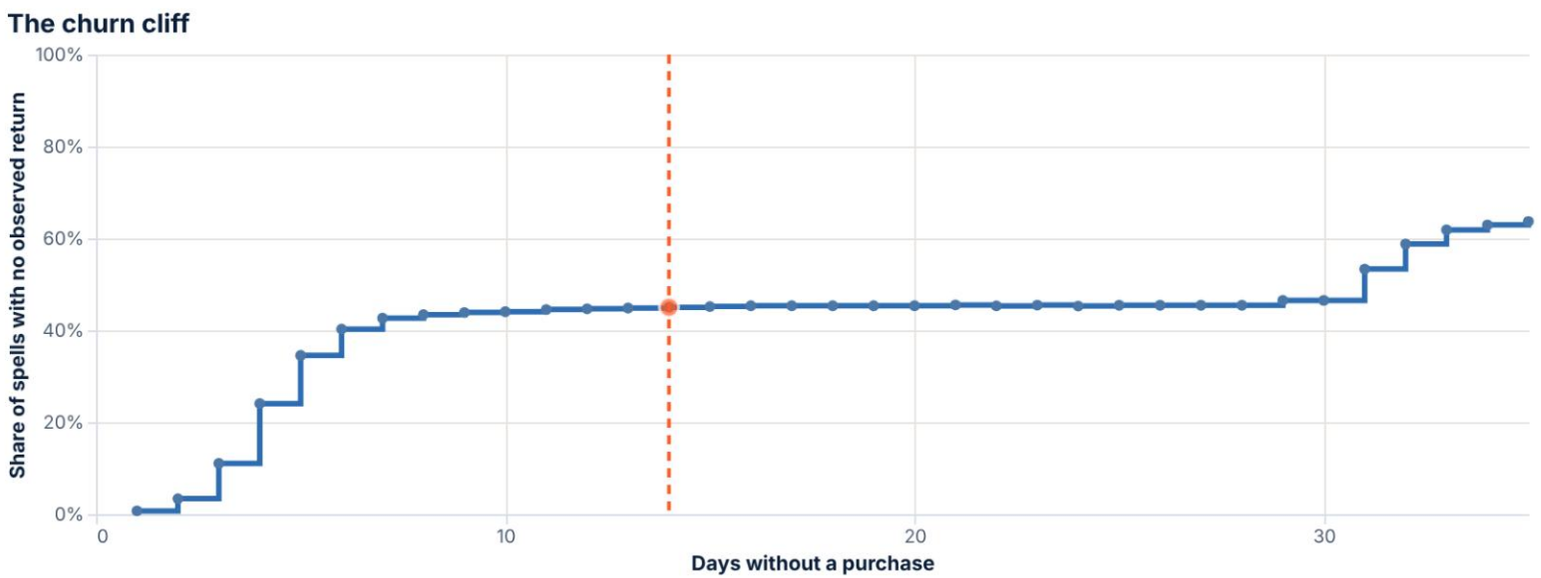
Metric	In person	Online
Active users	384	385
Mean ticket	375 NOK	1,007 NOK
Median ticket	241 NOK	395 NOK
Take rate	1.29%	**1.95%**

Online represents **14.2%** of purchases, but **30.7%** of GMV and **40.1%** of service revenue. Its take rate is **1.95%**, compared with **1.29%** in person.

DEFINING RETENTION

What does it mean to keep coming back?

The churn cliff



No observed return by 14 days

45%

Across 271 purchase spells

Users active within the 28-day window

299

Last purchase within 28 days of year-end

RETENTION FRAMEWORK

Three layers of “keep coming back”

01 / SURVIVAL

Recent activity

Purchased within the selected survival window.

02 / CADENCE

Expected frequency

Weekly in-person and monthly online activity.

03 / SPEND

Customer value

Purchase frequency and GMV.

Based on the RFM framework

JAN–APR · BEFORE VIP LAUNCH

Future VIPs were already more engaged

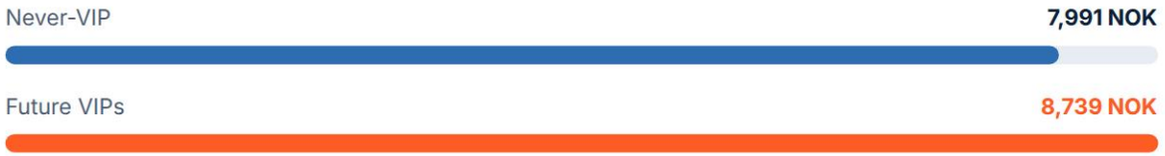
Before VIP launched, users who later enrolled were already spending more and purchasing more often than users who never enrolled.

AVERAGE WEEKLY GMV · JAN–APR · BEFORE VIP LAUNCH

1.09×

Future VIP average vs. Never-VIP

N=189 vs. N=196 · Welch t-test p = 0.191

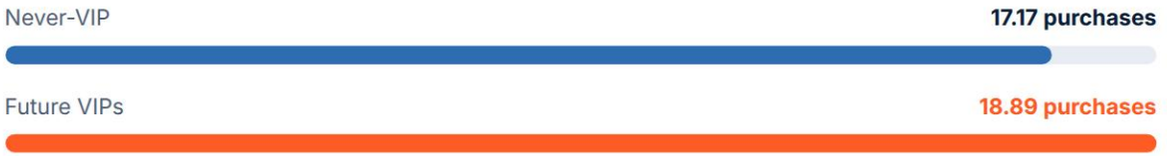


AVERAGE WEEKLY PURCHASES · JAN–APR · BEFORE VIP LAUNCH

1.10×

Future VIP average vs. Never-VIP

N=189 vs. N=196 · Welch t-test p = 0.084

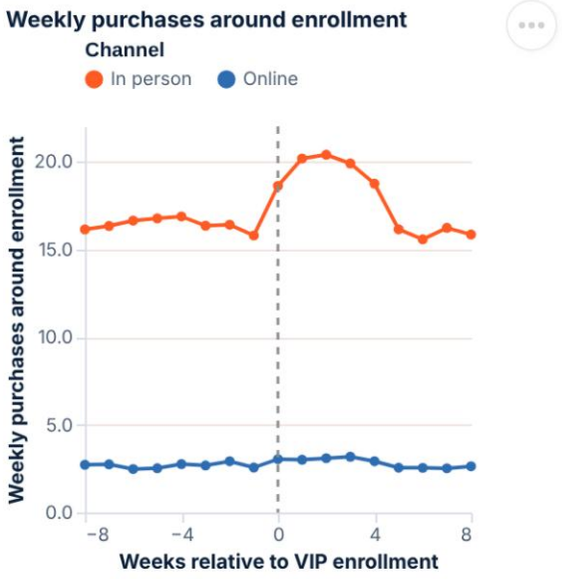


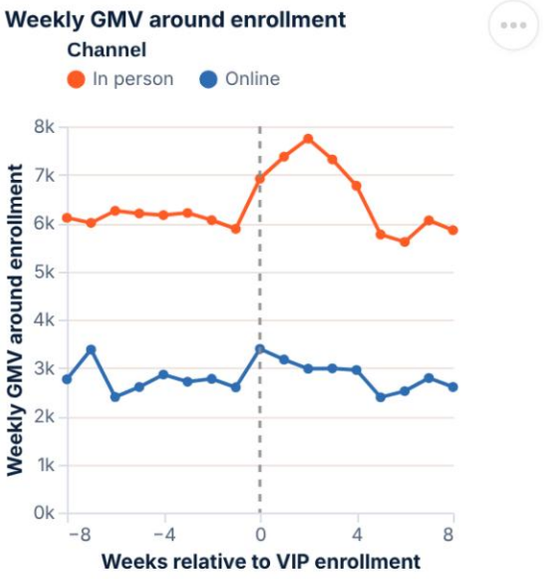
How to interpret this: This pre-existing difference is evidence of selection bias: the full-year gap cannot be interpreted as the effect of VIP membership alone.

Q1 · DO PEOPLE WHO JOIN VIP BUY MORE?

VIP enrollment coincides with higher in-person activity

Activity for the selected cohort before and after enrollment. The dashed vertical line marks the enrollment week.





BEFORE VS. AFTER ENROLLMENT

Metric	Before	After	Change	p-value
Purchases / week	19.1	20.7	+8.4%	0.000
In-person GMV / week	6,108 NOK	6,560 NOK	+7.4%	0.000
Online GMV / week	2,758 NOK	2,798 NOK	+1.4%	0.752
Total GMV / week	8,866 NOK	9,358 NOK	+5.5%	0.014

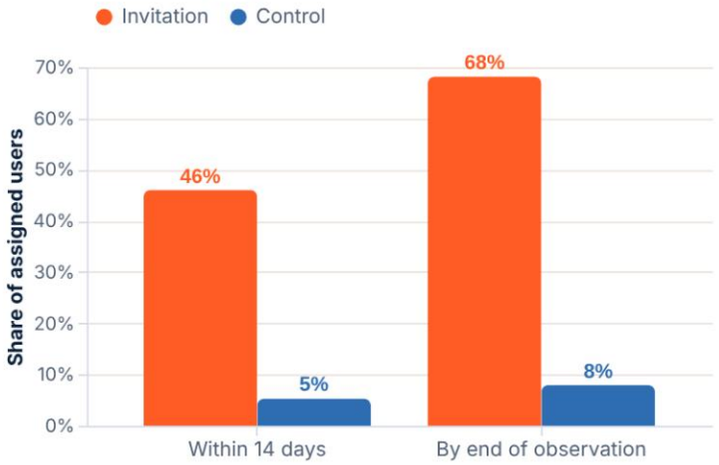
How to read this (N=177): Each user's average weekly activity before enrollment is compared with their average after enrollment. The change is an association, not proof that VIP caused it, because users choose when to join.

Q2 · DOES INVITING PEOPLE TO JOIN VIP HELP IN MAKING THEM COME BACK?

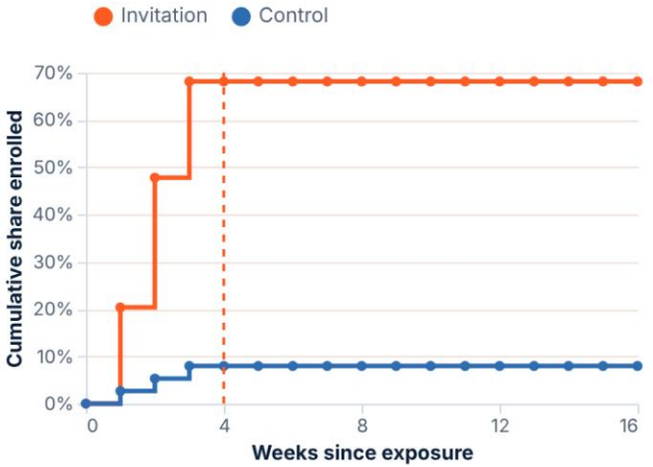
The invitation increased VIP enrollment

Compare enrollment between assigned experiment arms, then follow how the gap develops over time.

Enrollment rate by experiment arm



Cumulative enrollment over time



BASELINE CHARACTERISTICS

Baseline	Invitation	Control	p-value
June transactions (selection variable)	81.65	81.79	0.980
Pre-exposure purchases / week	18.14	18.05	0.940

Invited and Control arms had near-identical June purchase frequency prior to assignment (p = 0.903), confirming balanced randomization.

CUMULATIVE ENROLLMENT · WEEK 4

68.1% invitation vs. **8.0% control**

Difference: +60.2 percentage points

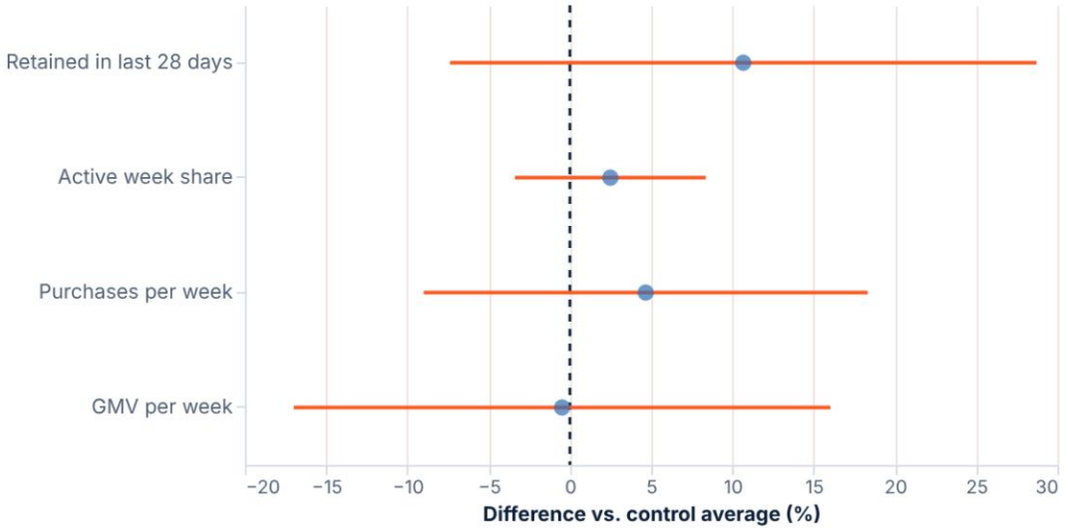
Ever-enrollment difference across the observation period: +60.2 pp (p < 0.001).

Q2 · DOES INVITING PEOPLE TO JOIN VIP HELP IN MAKING THEM COME BACK?

The invitation did not show a clear change in retention or spend

Compare post-exposure activity and spending between invitation and control users.

Invitation vs. control · 16-week window



OUTCOME ESTIMATES

Metric	Control	Invitation	Diff. [95% CI]	p
Retained in last 28 days	66.4%	73.5%	+10.7% [-7.4, +28.7]	0.248
Active week share	88.9%	91.1%	+2.5% [-3.4, +8.4]	0.407
Purchases per week	16.7	17.5	+4.7% [-9.0, +18.3]	0.504
GMV per week	7,945 NOK	7,907 NOK	-0.5% [-17.0, +16.0]	0.955

INTERPRETATION

All 95% confidence intervals include zero, so we cannot conclude the invitation changed retention or spending.

Points show estimated differences; lines show 95% confidence intervals. June-frequency groups use pre-experiment activity.

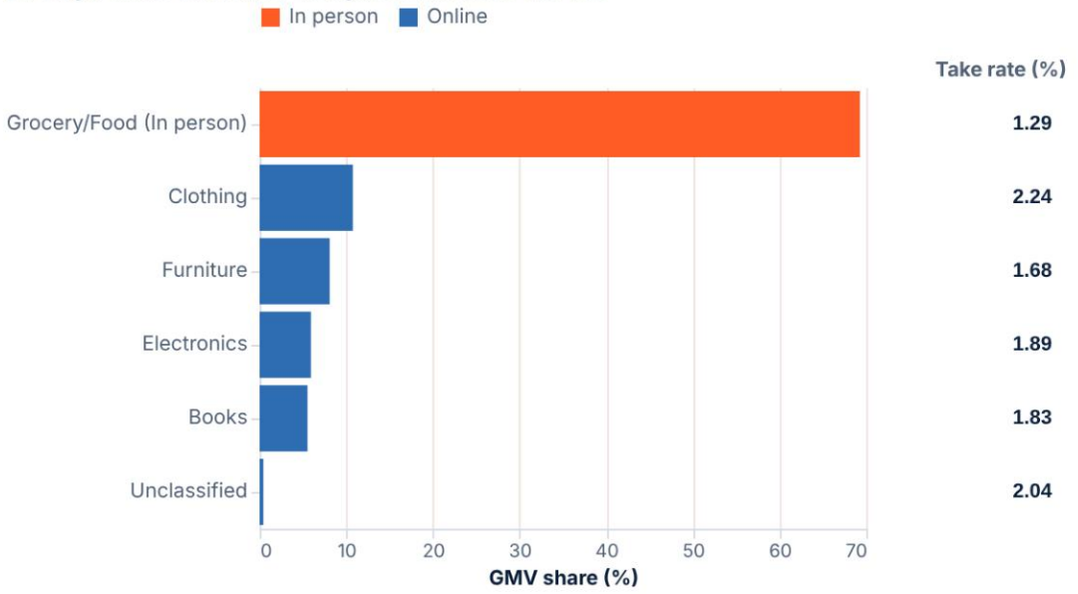
Q3 · RECOMMENDATIONS & NEXT STEPS

Turn VIP membership into an active habit

Sector GMV and take rates can help identify where online cross-sell may be worth testing.

GMV share by sector

Bar length shows GMV share; the right column shows take rate



GMV share describes transaction value mix; take rate is shown as a label and is not encoded by bar length.

IDEAS TO TEST

1. Replace batch signup invites with a 14-day inactivity trigger
2. Test online checkout point multipliers to encourage online purchases.

ILLUSTRATIVE ONLINE CROSS-SELL SCENARIO

What if each VIP made **1.0 extra online purchases per month?**

1 · Extra purchases / year	2 · Incremental online GMV	3 · Estimated fee revenue	4 · Point cost
2,340	**NOK 2,359,799**	**NOK 46,025**	**– NOK 11,799**
VIPs × 12 × 1.0	Purchases × 1,008 NOK average ticket	GMV × 1.95% take rate	GMV × 50 bps assumed cost

ESTIMATED FEE REVENUE LESS POINT COST

Fee revenue – point cost

NOK 34,226

Illustrative only. Assumes 195 VIP users (2025 total). Take rate is average of all online take rates.

Q3 · RECOMMENDATIONS & NEXT STEPS

What we left for later

Focus the analysis on reliable, decision-ready evidence. Revisit these areas when the data can support stronger conclusions.

DEFERRED ANALYSES

01 · CARD-LEVEL ANALYSIS

cards.csv

Why deferred
Card status is only known at the 2026-01-01 cutoff, not historical purchase time. Optional table, so not investigated further.

Revisit when: card history (issued, blocked, cancelled) exists.

02 · AGE SEGMENTATION

16 age buckets

Why deferred
Splitting 419 users into 16 four-year bands leaves about 26 per band, and only ~7 per experiment arm — too few to compare reliably.

Revisit with: pooled years or wider age bands.

03 · DECEMBER RETENTION

December wave

Why deferred
Limited days of observation were available before the 2026-01-01 cutoff.

Revisit in: Q1 2026, once 16 weeks elapse.

Principle: Prioritize analyses that are interpretable now; revisit deferred questions when the data can support them.

APPENDIX · SOURCES, LINEAGE & AI WORKFLOW

Reproducible pipeline & human-in-the-loop AI workflow

All metrics, marts, and interactive visuals are generated reproducibly from immutable raw CSVs.

01 · CODE, NOTEBOOKS & PLOT ARCHIVE

GitHub Repository — vipps_da_interview_case

Full Analysis Workbook — notebooks/case_analysis.py

Interactive Slide Deck — notebooks/vip_presentation.py

Static Plot Gallery — plots/

02 · DATA LINEAGE & SEMANTIC MODEL

Raw Case Data `data/raw/` — transactions.csv, users.csv, merchants.csv, mcc.csv, experiment_exposure.csv, cards.csv (deferred)

Derived Marts `data/processed/` — df_user_spine.csv, df_exp_eval.csv, df_weekly_panel.csv

Semantic Layer `semantic/` — model.yml, marts.yml, erd.md

03 · TECH STACK & STATISTICAL METHODS

Compute & QA — Python, DuckDB, Pandas, NumPy, ruff, pytest

Visualization — Marimo + Altair (Vega-Lite)

Statistical Inference `scipy.stats` — Welch t-test, paired t-test, two-proportion z-test, complier-scaled ITT

04 · AI MODELS & HUMAN VERIFICATION

Planning & Analytical Partner — Gemini 4 Argon

Coding & Implementation Model — DeepSeek V4.1

Human-in-the-Loop Governance — every metric, cohort filter, and conclusion verified against raw data

Thanks for your time & questions are welcome!